Caltech-101 Classification Report: Classical Baseline, Transfer Learning, and Ablation Studies

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Abstract

This report summarizes training and evaluation results for Caltech-101 classification over 102 classes (including `BACKGROUND_Google`). We compare two classical baselines (HOG + Linear SVM, HOG + Engineered Features + LightGBM) against transfer-learning models (ResNet50, EfficientNet-B0, ViT-B/16), then analyze ablations on image size, augmentation, and optimizer. On the main test runs, ViT-B/16 achieves the best Top-1 accuracy (0.9461) and Macro-F1 (0.9272), followed by ResNet50 (0.9402, 0.9183) and EfficientNet-B0 (0.9213, 0.8955). Among classical methods, LightGBM improves over SVM (Top-1 0.5437 vs 0.5036), but both remain substantially below deep transfer-learning performance. Ablations show strong dependence on image resolution for CNNs, limited gains from the tested augmentation setup, and a large optimizer sensitivity for ResNet50 (Adam $\gg$ SGD in this setup).

1 Introduction

Caltech-101 is an important benchmark for multi-class object recognition, and it has been widely used to evaluate both classical and modern visual representation learning methods [1]. Traditional pipelines often combine hand-crafted Histogram of Oriented Gradients (HOG) descriptors [2] with Support Vector Machine (SVM) classifiers [3], while recent progress is dominated by deep architectures such as ResNet [4], EfficientNet [5], and Vision Transformer (ViT) [6].

In this work, we evaluate Caltech-101 classification under a unified experimental framework using all 102 classes, including `BACKGROUND_Google`, with a stratified split protocol (train/val/test = 6400/1372/1372 images, i.e., 70.0%/15.0%/15.0%). We implement a classical baseline (HOG + Linear SVM) and multiple transfer-learning deep models (ResNet50, EfficientNet-B0, and ViT-B/16), report Top-1 accuracy, Top-5 accuracy, Macro-F1, and Weighted-F1, and run ablation studies on image resolution, augmentation policy, and optimizer choice.

The purpose of this report is to provide a rigorous empirical comparison between classical machine learning models and deep learning models under the same split protocol, and to clearly characterize the performance differences between these two model families on Caltech-101.

2 Dataset

All experiments use Caltech-101 [1], with 9,144 images across 102 classes. We construct data splits with a two-stage stratified procedure over class labels: first, 70% training and 30% temporary data; second, the temporary set is split evenly into validation and test sets, yielding 6,400/1,372/1,372 images (70.0%/15.0%/15.0%). This preserves class proportions across splits and avoids sample overlap.

We intentionally keep `BACKGROUND_Google` as a valid class, rather than discarding it, to better reflect realistic recognition conditions where images may not correspond to a clean foreground object category. Including this class makes the task more challenging and more repre-

sentative, and it allows us to evaluate whether models can separate true object categories from heterogeneous background-like content.

Preprocessing is model-specific but standardized within each family. For the classical pipeline, each image is first converted to RGB to standardize mixed source formats into a consistent 3-channel input, then resized to 128×128 to keep feature dimensionality fixed, scaled to $[0, 1]$ for numerical stability, converted to grayscale so HOG focuses on intensity-gradient structure, and finally encoded with HOG features (9 orientations, 8×8 pixels per cell, 2×2 cells per block). For deep models, ResNet50 and EfficientNet-B0 are trained at 128×128, while ViT-B/16 is run at 224×224 to satisfy architecture input requirements; all deep inputs are normalized with ImageNet mean/std to match pretrained feature statistics, and training uses either deterministic resize without augmentation or RandomResizedCrop + horizontal flip + color jitter with augmentation for regularization, while validation/test always use deterministic resize and normalization for consistent evaluation.

3 Methods

3.1 HOG + Linear SVM

As a classical baseline, we employ a linear Support Vector Machine (SVM) on HOG representations. Model selection is performed on the validation split by maximizing Macro-F1 over the regularization parameter C, and the selected model is evaluated on the held-out test set.

3.2 HOG + Engineered Features + LightGBM

To provide a stronger non-linear classical comparator, we concatenate HOG features with nine engineered image statistics (width, height, aspect ratio, channel-wise RGB means, and channel-wise RGB standard deviations) and train a multiclass LightGBM classifier. Hyperparameters (`num_leaves`, `learning_rate`, `n_estimators`) are selected using validation Macro-F1.

3.3 Transfer-Learning Deep Models

We fine-tune three ImageNet-pretrained backbones and adapt each architecture to 102-way classification by replacing the final prediction head.

3.3.1 ResNet50

ResNet50 is a residual convolutional architecture in which skip connections enable stable optimization of deep networks. We replace the final fully connected layer with a 102-class classifier and fine-tune the model with cross-entropy, selecting the checkpoint that achieves the best validation Macro-F1.

3.3.2 EfficientNet-B0

EfficientNet-B0 is a compact convolutional model that balances depth, width, and resolution through compound scaling. We replace its classifier layer with a 102-class head and fine-tune it under the same training/selection criterion, retaining the best validation Macro-F1 checkpoint.

3.3.3 ViT-B/16

ViT-B/16 is a transformer-based vision architecture that models global image context via self-attention over patch tokens. We replace the transformer head with a 102-class classifier and fine-tune the model with cross-entropy, using the best validation Macro-F1 checkpoint for test evaluation.

3.4 Training Protocol

For all deep models, training lasts 12 epochs with a two-stage freeze–unfreeze strategy: the backbone is frozen for the first 2 epochs and then unfrozen for full fine-tuning. ResNet50 and EfficientNet-B0 use batch size 32, initial learning rate 3×10^{-4}, and post-unfreeze learning rate 1×10^{-4} at 128×128 resolution, whereas ViT-B/16 uses batch size 16, initial learning rate 1×10^{-4}, and post-unfreeze learning rate 5×10^{-5} at 224×224; in all deep runs, Adam is the default optimizer with weight decay 1×10^{-4}.

For the classical models, HOG extraction uses size 128, 9 orientations, 8×8 pixels per cell, and 2×2 cells per block. The linear SVM is trained with `max_iter`=1000 and validation tuning over C (configured grid: [0.1]). The LightGBM baseline is intentionally configured for fast execution with a single setting (`num_leaves`=31, `learning_rate`=0.1, `n_estimators`=100, `subsample`=0.8, `colsample_bytree`=0.8), and model selection remains based on validation Macro-F1; GPU mode can be requested in the training script, with automatic CPU fallback if GPU acceleration is unavailable in the LightGBM build.

4 Experiment Setup

4.1 Augmentation

To isolate the effect of data augmentation, we evaluate two training pipelines for deep models while keeping the remaining settings fixed. The first pipeline applies only deterministic resizing to the target input resolution. The second pipeline applies stochastic augmentation during training (random resized crop, horizontal flip, and color jitter), which acts as regularization by increasing appearance variability. For both pipelines, validation and test images are processed with deterministic resizing and normalization to ensure a controlled and comparable evaluation protocol.

4.2 Optimization

Deep models are optimized with cross-entropy loss under a freeze–unfreeze transfer-learning schedule: the pretrained backbone is frozen in early epochs and then unfrozen for end-to-end fine-tuning. Adam is used as the default optimizer in the main comparison runs, with weight decay applied as regularization. To explicitly assess optimizer effects, we also run a controlled comparison in which ResNet50 is trained twice under identical image size and augmentation settings, once with Adam and once with SGD.

4.3 Model Selection

For deep models, checkpoint selection is based on validation Macro-F1, and the best checkpoint is used for final test evaluation. We prefer Macro-F1 over overall accuracy because the class distribution is imbalanced; accuracy can be dominated by frequent classes, whereas Macro-F1 gives equal weight to each class and therefore better reflects balanced recognition quality across all 102 categories.

4.4 Ablation Protocol

- **A1 (Image Resolution):** ResNet50 and EfficientNet-B0 are evaluated at 64 and 128 input resolution, with all other settings fixed. ViT-B/16 remains at 224 due to architecture constraints.
- **A2 (Augmentation Effect):** each deep model is trained with and without stochastic augmentation at fixed resolution and optimizer settings.

- **A3 (Optimizer Effect):** ResNet50 is trained with Adam and SGD under matched settings to isolate optimizer impact.

5 Results

5.1 Overall Performance

Table 1: Main test results from selected experiment runs.

Model	Top-1 Acc	Top-5 Acc	Macro-F1	Weighted-F1
Classical SVM	0.5036	0.6334	0.3043	0.4741
Classical LightGBM	0.5437	0.6968	0.3032	0.5376
EfficientNet-B0	0.9213	0.9869	0.8955	0.9205
ResNet50	0.9402	0.9942	0.9183	0.9402
ViT-B/16	**0.9461**	0.9913	**0.9272**	**0.9459**

Table 1 addresses the central objective of this report: a controlled comparison between classical and transfer-learning approaches under a shared split protocol and evaluation suite. The results show a clear hierarchy in which deep transfer-learning models substantially outperform classical baselines, with ViT-B/16 achieving the highest Top-1 accuracy and Macro-F1. LightGBM improves over linear SVM in Top-1 and Weighted-F1, but this gain is not sufficient to close the large gap to pretrained deep representations, reinforcing the practical motivation for transfer learning on this benchmark.

Table 2: Classical model training outcomes (selected by validation Macro-F1).

Model	Hyperparameters	Val Macro-F1
HOG + Linear SVM	$C = 0.1$	0.2819
HOG + Engineered Features + LightGBM	leaves=31, lr=0.1, trees=100	0.3198

Table 2 summarizes the classical-model training configuration selected by validation Macro-F1. Under the current fast-baseline setup, LightGBM attains a higher validation Macro-F1 than linear SVM, which is consistent with its stronger non-linear decision capacity and its improved Top-1 test accuracy in Table 1.

5.2 Per-Class Results

To quantify class-wise behavior, we report per-class accuracy for each class c as

$$\mathrm{Acc}_c = \frac{\#\{\text{correct predictions in class } c\}}{N_c} = \frac{\mathrm{TP}_c}{N_c},$$

where N_c is the number of test samples in class c. We summarize these values by their arithmetic mean across classes:

$$\overline{\mathrm{Acc}}_{\text{class}} = \frac{1}{C}\sum_{c=1}^{C} \mathrm{Acc}_c, \quad C = 102.$$

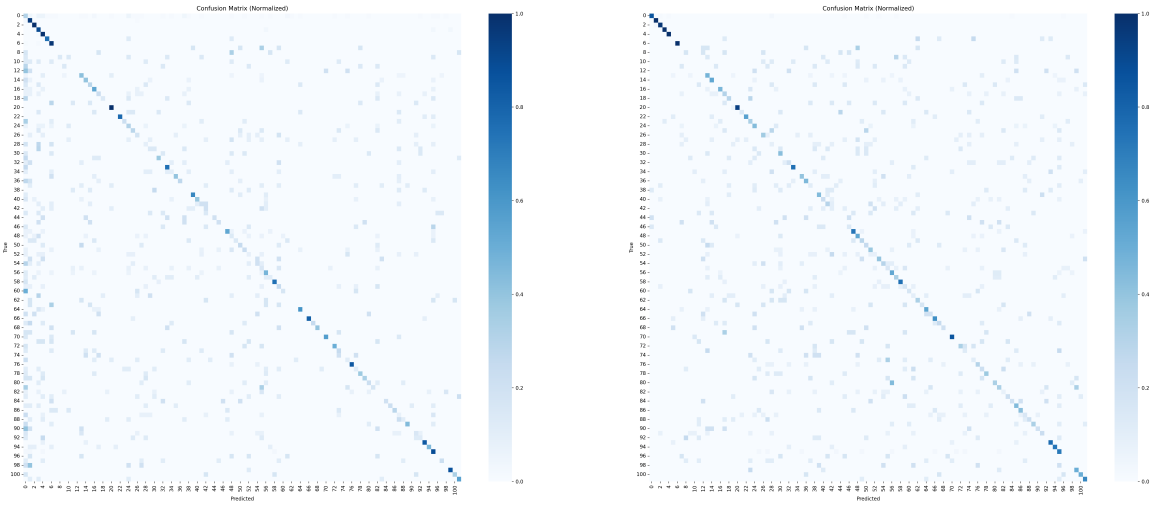
This analysis is necessary because the dataset is class-imbalanced; aggregate accuracy can mask poor performance on low-frequency or visually difficult categories.

Average per-class accuracy across the 102 classes is:

- Classical SVM: 0.2893

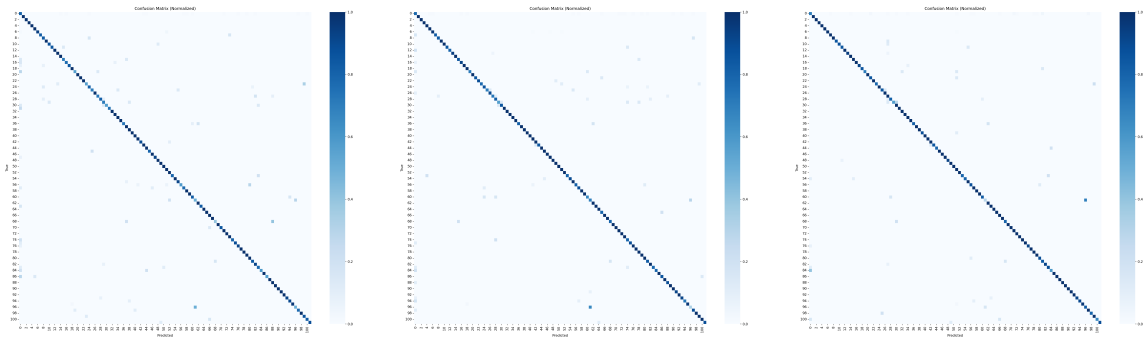
- Classical LightGBM: 0.3076
- EfficientNet-B0: 0.8977
- ResNet50: 0.9222
- ViT-B/16: 0.9350

Additionally, 37 classes are predicted perfectly (accuracy = 1.0) by all three deep models.



(a) HOG + Linear SVM (b) HOG + Engineered Features + LightGBM

Figure 1: Normalized confusion matrices for the two classical models.



(a) EfficientNet-B0 (b) ResNet50 (c) ViT-B/16

Figure 2: Normalized confusion matrices for the three transfer-learning models.

The confusion matrices are consistent with the metric-level findings. Both classical models exhibit broader off-diagonal spread, indicating frequent cross-class confusion and weak discrimination for many categories; LightGBM slightly sharpens the diagonal relative to SVM, matching its higher Top-1 accuracy. In contrast, all deep models produce strongly diagonal matrices, reflecting robust class separation. Among them, ViT-B/16 and ResNet50 show the cleanest diagonal concentration, while EfficientNet-B0 remains strong but with relatively more residual confusion on hard minority classes.

6 Ablation Study

We conduct a targeted ablation study to isolate the contribution of three implementation factors in the deep-learning pipeline: input resolution, augmentation policy, and optimizer choice. Each ablation changes one factor while holding the others fixed, so that performance differences can be interpreted causally rather than as the result of multiple simultaneous configuration changes.

6.1 A1: Image Size (64 vs 128)

Table 3: A1 image-size ablation (Top-1 accuracy and Macro-F1).

Model	Top-1@64	Top-1@128	ΔTop-1	Macro-F1@64	Macro-F1@128	ΔMacro-F1
ResNet50	0.8739	0.9402	+0.0663	0.8318	0.9183	+0.0864
EfficientNet-B0	0.8061	0.9213	+0.1152	0.7427	0.8955	+0.1528
ViT-B/16	0.9461	0.9461	+0.0000	0.9272	0.9272	+0.0000

Increasing resolution from 64 to 128 substantially improves both Top-1 and Macro-F1 for CNN backbones, with the largest gain on EfficientNet-B0. The larger Macro-F1 gains (especially +0.1528 for EfficientNet-B0) indicate that higher resolution benefits not only overall accuracy but also balanced performance across classes. ViT-B/16 remains unchanged because its implementation is constrained to 224 input resolution in both settings.

6.2 A2: Augmentation (Without vs With)

Table 4: A2 augmentation ablation (Top-1 accuracy and Macro-F1).

Model	Top-1 WO	Top-1 W	ΔTop-1	Macro-F1 WO	Macro-F1 W	ΔMacro-F1
ResNet50	0.9410	0.9402	-0.0007	0.9196	0.9183	-0.0014
EfficientNet-B0	0.9359	0.9213	-0.0146	0.9065	0.8955	-0.0110
ViT-B/16	0.9592	0.9461	-0.0131	0.9400	0.9272	-0.0128

Note: WO = without augmentation, W = with augmentation. In this configuration, stochastic augmentation degrades both Top-1 and Macro-F1 for all three deep models. The consistent negative deltas suggest that the selected augmentation strength/type is not well matched to this training budget and dataset regime, and may require retuning rather than being assumed beneficial by default.

6.3 A3: Optimizer (Adam vs SGD on ResNet50)

Table 5: A3 optimizer ablation on ResNet50 (Top-1 accuracy and Macro-F1).

Optimizer	Top-1 Acc	ΔTop-1 vs Adam	Macro-F1	ΔMacro-F1 vs Adam
Adam	0.9402	0.0000	0.9183	0.0000
SGD	0.8411	-0.0991	0.7472	-0.1710

Adam clearly outperforms SGD under the current hyperparameter settings, with a large Top-1 gap (-0.0991) and an even larger Macro-F1 gap (-0.1710) for SGD. The larger Macro-F1 drop

implies that SGD in this setup disproportionately harms class-balanced performance, not only average accuracy.

7 Observation and Discussion

7.1 Deep Models vs Classical Baselines

Deep transfer-learning models substantially outperform the classical baselines, with ViT-B/16 achieving the best overall performance (Top-1 0.9461, Macro-F1 0.9272). This gap indicates that pretrained deep representations are the most reliable option for this 102-class setting, especially when balanced class-level performance is a priority.

7.2 SVM vs LightGBM Baseline Behavior

LightGBM improves Top-1 and Weighted-F1 over linear SVM, indicating better non-linear fitting capacity for global decision boundaries. However, its Macro-F1 remains low and close to SVM, implying that classical pipelines still struggle to generalize across rare and difficult categories.

7.3 Implications from Resolution and Optimizer Ablations

The ablations show that image resolution and optimizer selection are high-impact controls in this pipeline. In practical terms, moving from 64 to 128 strongly improves CNN performance, and Adam is clearly more suitable than SGD under the tested ResNet50 hyperparameters.

7.4 Implications from Augmentation Ablation

The tested augmentation policy yields consistent metric decreases across all three deep models. This suggests that augmentation should be treated as a tunable design variable rather than a default improvement, and its strength/composition should be re-optimized for this dataset and training budget.

7.5 Failure Modes

A consistent failure pattern is concentrated on classes with small test support and/or high visual ambiguity. For example, classes such as `lotus`, `crocodile`, and `snoopy` remain among the hardest even for strong deep models, while several low-support classes reach near-zero accuracy under classical models. The confusion matrices also indicate that heterogeneous categories (including `BACKGROUND_Google`) are more prone to cross-class confusion, which suggests sensitivity to cluttered scenes, intra-class variability, and limited sample diversity in minority classes.

7.6 Limitations

This study is limited by a single split protocol and a fixed seed in the reported main runs, so variance across random initializations and split realizations is not fully characterized. Hyperparameter exploration is also intentionally constrained for some settings (notably the fast LightGBM baseline), which prioritizes runtime over peak performance. In addition, ViT-B/16 input size is architecture-constrained in this codebase, and augmentation exploration is restricted to one policy family; broader augmentation and optimization sweeps may yield different conclusions. Finally, results are reported on a single dataset benchmark, so cross-dataset generalization is not evaluated.

8 Reproducibility

To support reproducibility, the full source code, training scripts, configuration files, and report artifacts are available in the project repository:

`https://github.com/G6KlayWang/Caltech_101`

Users can reproduce the main experiments by following the documented setup and running the provided shell entrypoints for data preparation, model training, and evaluation.

References

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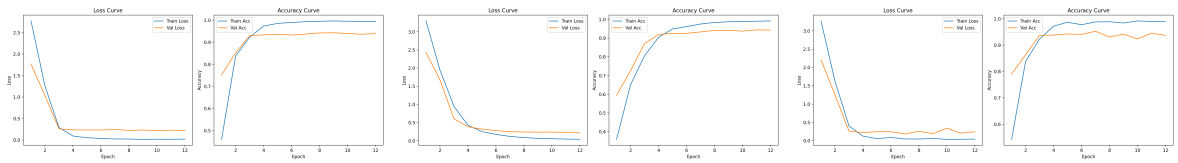
A Appendix

A.1 Training Curves for Deep Models



(a) ResNet50 (b) EfficientNet-B0 (c) ViT-B/16

Figure 3: Training curves for the three main deep-learning runs.

A.2 Top-5 and Bottom-5 Per-Class Accuracy

A.2.1 HOG + Linear SVM

Table 6: Bottom-5 classes for HOG + Linear SVM.

Class	Accuracy	Test Count
binocular	0.0000	5
inline_skate	0.0000	5
metronome	0.0000	5
platypus	0.0000	5
strawberry	0.0000	5

Table 7: Top-5 classes for HOG + Linear SVM.

Class	Accuracy	Test Count
Faces_easy	1.0000	66
car_side	1.0000	18
Motorbikes	0.9917	120
airplanes	0.9750	120
Faces	0.9545	66

A.2.2 HOG + Engineered Features + LightGBM

Table 8: Bottom-5 classes for HOG + Engineered Features + LightGBM.

Class	Accuracy	Test Count
binocular	0.0000	5
garfield	0.0000	5
gerenuk	0.0000	5
platypus	0.0000	5
wild_cat	0.0000	5

Table 9: Top-5 classes for HOG + Engineered Features + LightGBM.

Class	Accuracy	Test Count
Motorbikes	1.0000	120
airplanes	1.0000	120
Faces	1.0000	66
Leopards	1.0000	30
Faces_easy	0.9697	66

A.2.3 EfficientNet-B0

Table 10: Bottom-5 classes for EfficientNet-B0.

Class	Accuracy	Test Count
octopus	0.4000	5
water_lilly	0.5000	6
lotus	0.5000	10
chair	0.5556	9
cannon	0.5714	7

Table 11: Top-5 classes for EfficientNet-B0.

Class	Accuracy	Test Count
Motorbikes	1.0000	120
Faces	1.0000	66
Leopards	1.0000	30
bonsai	1.0000	19
car_side	1.0000	18

A.2.4 ResNet50

Table 12: Bottom-5 classes for ResNet50.

Class	Accuracy	Test Count
water_lilly	0.3333	6
crocodile	0.5714	7
lobster	0.6667	6
lotus	0.7000	10
cannon	0.7143	7

Table 13: Top-5 classes for ResNet50.

Class	Accuracy	Test Count
Motorbikes	1.0000	120
Faces	1.0000	66
Faces_easy	1.0000	66
Leopards	1.0000	30
car_side	1.0000	18

A.2.5 ViT-B/16

Table 14: Bottom-5 classes for ViT-B/16.

Class	Accuracy	Test Count
lotus	0.3000	10
crocodile	0.5714	7
snoopy	0.6000	5
wrench	0.6667	6
BACKGROUND_Google	0.7000	70

Table 15: Top-5 classes for ViT-B/16.

Class	Accuracy	Test Count
Motorbikes	1.0000	120
Faces	1.0000	66
Faces_easy	1.0000	66
Leopards	1.0000	30
car_side	1.0000	18